COVENTRY UNIVERSITY | BSC (HONS) COMPUTING

OPTIMIZING COMPUTER VISION FOR TRANSPARENT PLASTIC BOTTLE CLASSIFICATION

VT6001CEM Individual Project Presentation

STUDENT
CHAN PUI YAN NEVALLE
ID: 257037727

1ST SUPERVISOR
CANDY WONG

2ND MARKER
MAGGIE LEE

HONG KONG
PLASTIC WASTE PROBLEM

2,120

TONS OF PLASTIC WASTE DAILY

Consumer Confusion:

Citizens struggle to sort visually identical transparent plastics (e.g., PET vs. PVC), contaminating recycling batches.

Technical Invisibility:

Standard AI detectors fail on transparent bottles because embossed Resin Identification Codes (RIC) have **zero color contrast** with the background body.

SYSTEM ARCHITECTURE, TECHNOLOGY STACK

INPUT CAPTURE

Smartphone photo captured and upload with targeted side-lighting to cast shadows on embossed RIC text.

OPENCV PREPROCESSING

CLAHE enhances local contrast, followed by Canny Edge Detection to convert invisible relief into white edge maps.

YOLOV8 CLASSIFIER

Preprocessed 3-channel edge map is fed into a lightweight YOLOv8 Nano model for material classification.

WEB UI FEEDBACK

Next.js web application delivers real-time expo camera detection with sub-second inference time (<3s).

FASTAPI BACKEND

High-performance backend API serving **YOLOv8 Nano** inference engine and **OpenCV** pipelines (CLAHE + Canny).

NEXT.JS & EXPO

Responsive Next.js client featuring Expo Camera API integration for seamless smartphone camera stream capturing.

VERCEL & CLOUDFLARE

Vercel serverless web application hosting paired with Cloudflare Tunnel for secure, low-latency API backend bridging.

ROBOFLOW MLOPS

Centralized platform for dataset hosting, manual polygon/box annotations, tag versioning (`pet1, pp5`), and augmentations.

MATERIAL SELECTION: PET (1) VS PP (5)

1. MARKET DOMINANCE IN HONG KONG

- **PET (Code 1):** Highest volume single-use beverage containers (Vitasoy, Watson's, Coca-Cola). High monetary recycling value.
- **PP (Code 5):** Primary resin for takeaway meal boxes, drink caps, and microwave containers.
- **Contamination Risk:** Mixing PET and PP destroys recycled resin batch quality in Hong Kong recycling facilities.

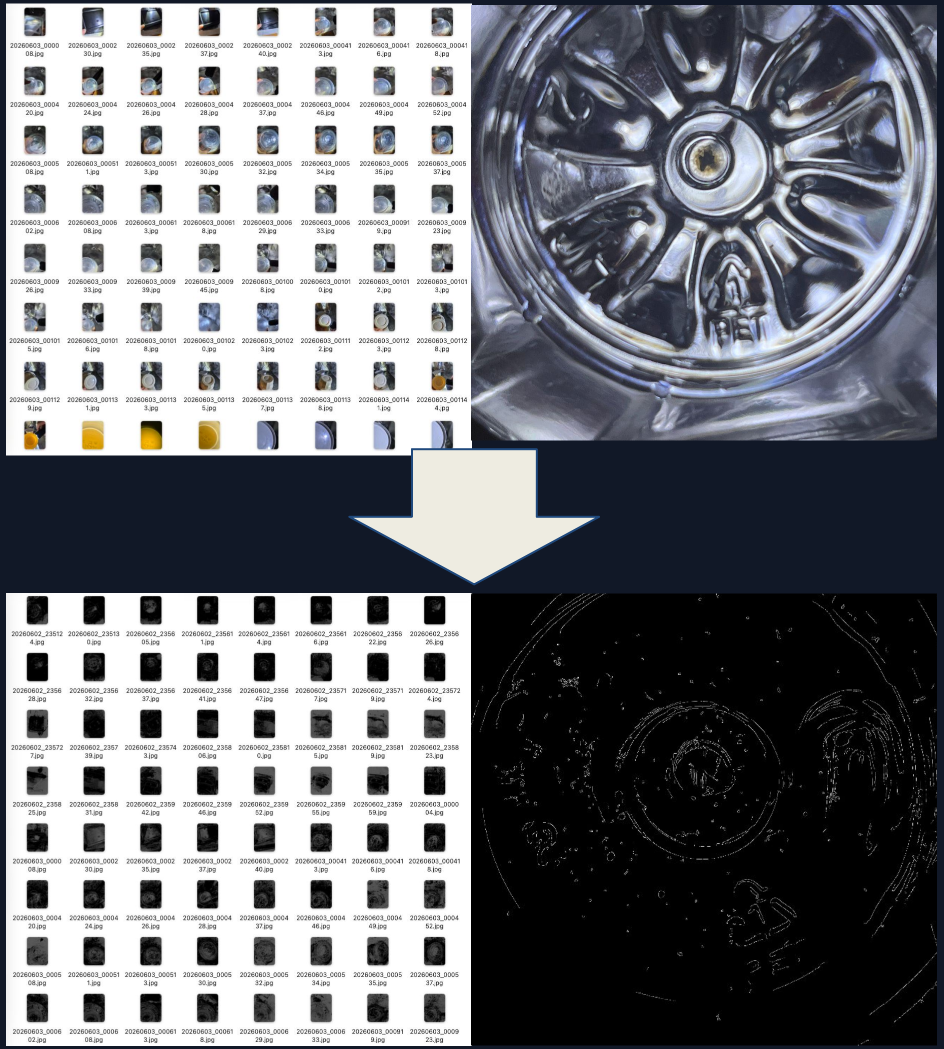
2. POLAR OPPOSITE OPTICAL PROFILES

- **PET (Code 1):** Ultra-transparent, thin walls, high specular reflections, and intricate petalloid rib geometry (high specular noise).
- **PP (Code 5):** Translucent / cloudy, thick walls, matte surface finish with diffuse light scattering (high texture noise).
- **Pipeline Validation:** Proves our Preprocessing-First pipeline scales across the full optical spectrum of consumer plastics.

HK LOCALIZED BENCHMARK DATASET

To rigorously evaluate models on local waste streams, an Hong Kong bottle dataset was constructed and categorized into sub-variants:

RIC Code	Sub-Variant	Sample Count
PET (RIC Code 1)	bottle raw images	30
	bottle images + preprocessed	30
	bottle images + preprocessed + data argumented	105
PP (RIC Code 5)	bottle raw image	103
	bottle images + preprocessed + data argumented	309

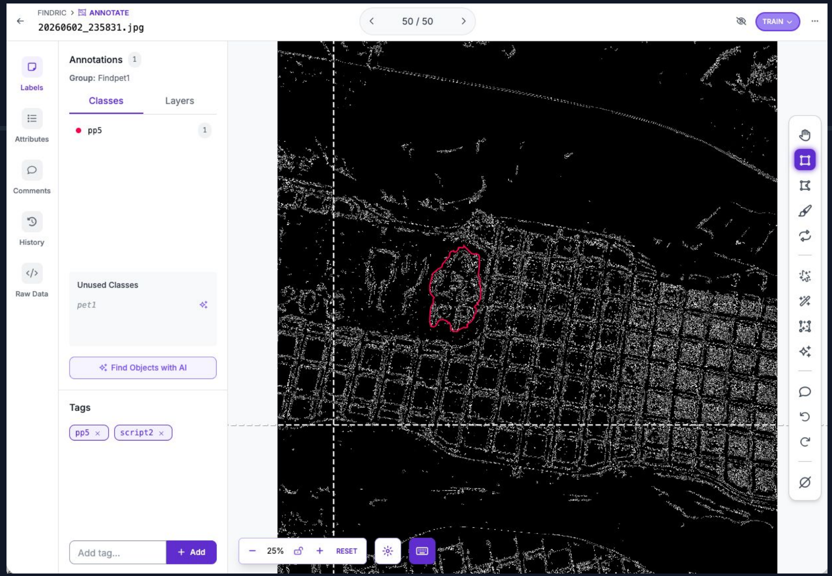


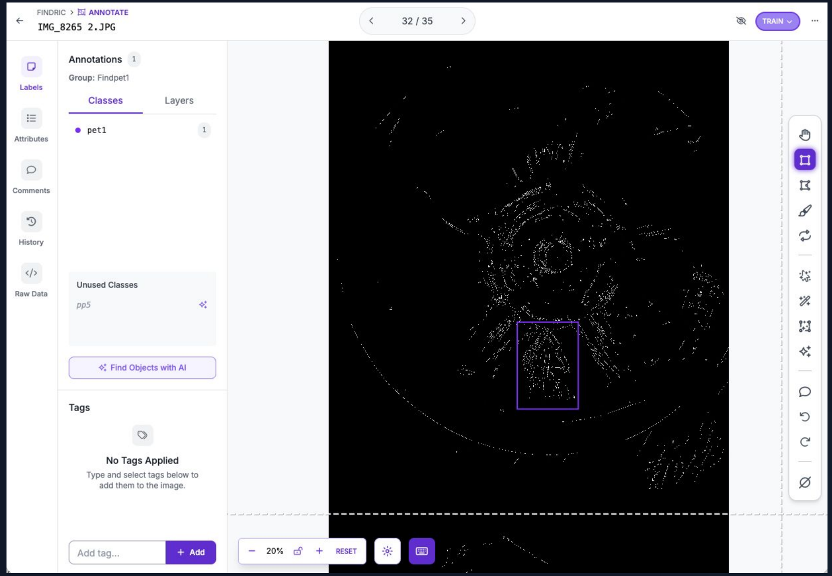
ROBOFLOW ANNOTATION

EDGE MAP LABELING STRATEGY

Annotating directly on preprocessed Canny edge maps revealed distinct material labeling requirements:

- **Bounding Box Labels (pet1):** Applied on clear PET bases. Precise box isolation is required to exclude surrounding concentric petalloid rib rings.
- **Polygon Segmentation (pp5):** Essential for PP 5 containers to isolate the RIC code from heavy square waffle/mesh background grid patterns.
- **Pipeline Version Control:** Dataset classes & tags (e.g pp5 pet1) track preprocessing script iterations directly in Roboflow.





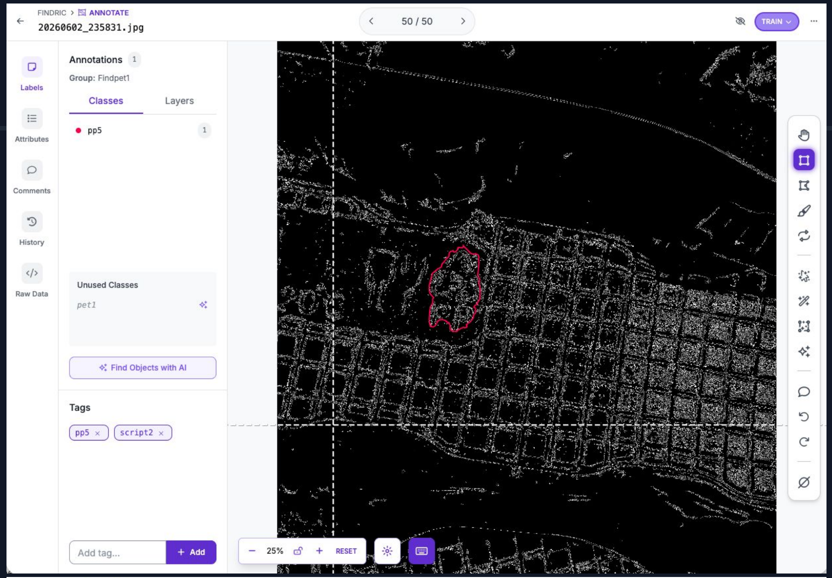
Findings: Loose bounding boxes capture background structural geometry (e.g., waffle grids and petalloid ribs) as false positive features. Tight, symbol-only annotations are critical to prevent YOLOv8 convolution weights from learning container mold structures.

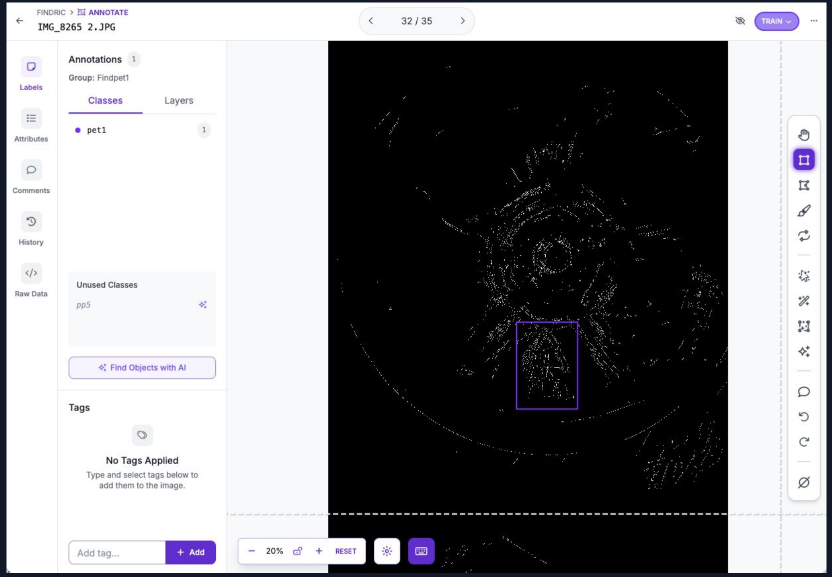
ROBOFLOW ANNOTATION

SOLVING MOLD STRUCTURAL NOISE

Annotating directly on Canny edge maps revealed critical MLOps findings:

- ❌ **The Loose Box Trap:** Standard loose bounding boxes capture surrounding container geometry (waffle grid meshes & petalloid ribs) as false features.
- **PET 1 Strategy:** Tight bounding box isolation is mandatory to exclude concentric petalloid rib rings from learning weights.
- **PP 5 Strategy:** Shifted to **tight polygon segmentation** to physically decouple the RIC triangle from heavy background grid patterns.
- **MLOps Versioning:** Embedded dataset tags (pp5 pet1) to track preprocessing hyperparameter iterations directly in Roboflow.





OPENCV FEATURE SOLIDIFICATION

MAKING "PHANTOM TEXT" VISIBLE

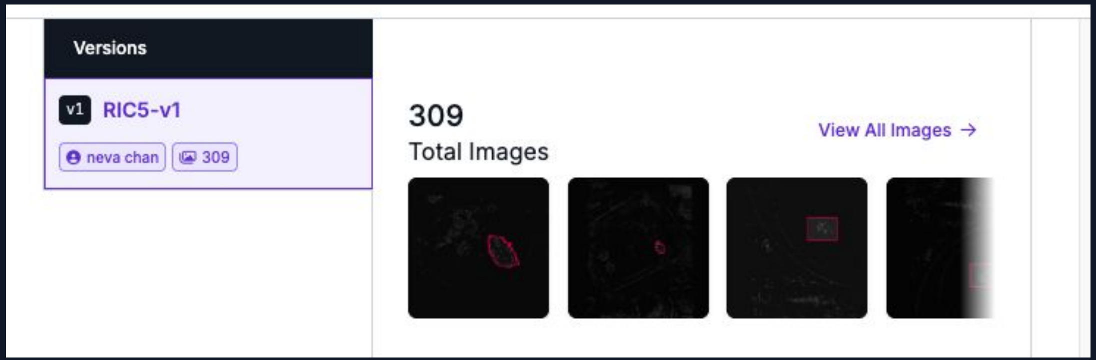
A Preprocessing-First Engineering Pipeline:

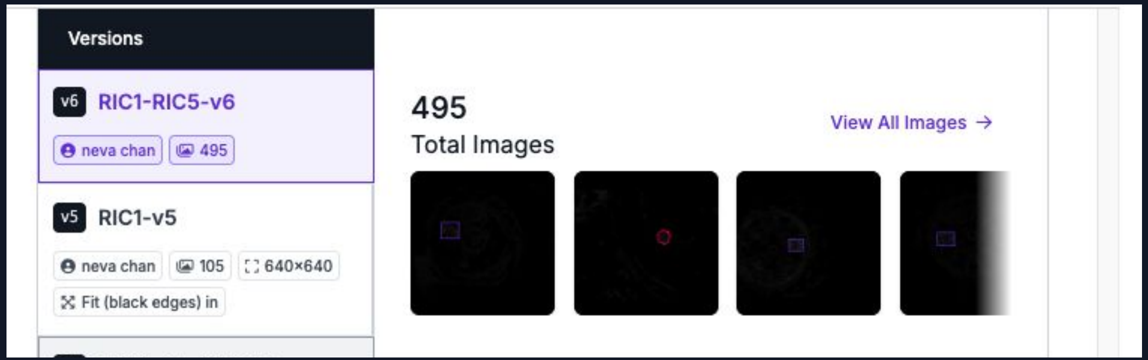
- **3-Channel Conversion:** Transforms output into RGB (`cv2.COLOR_GRAY2RGB`) to maintain structural compatibility with YOLOv8 backbone.
- **CLAHE Enhancement:**
  Adapts local contrast (`clipLimit=3.0`, `tileGridSize=8×8`) to accentuate faint embossed shadows on curved surfaces.
- **Canny Edge Extraction:**
  Detects sharp gradient boundaries, converting transparent relief into sharp geometric lines.

EXPERIMENTAL SETUP & MODEL VARIANTS

Four model pipelines were trained and benchmarked across the test dataset under controlled parameters:

Model Variant Name	Input Image Format	Specialized Class Focus	Key Engineering Purpose
YOLOv8 Baseline	Raw RGB Camera Photos	Off-the-shelf General	Evaluates standard AI zero-shot capability on transparent objects
RIC1-v5 Model	CLAHE + Canny Edge Maps	PET (Code 1) Specialized	Measures edge enhancement gain on clear shallow embossed PET
RIC5-v1 Model	Gaussian + Canny Edge Maps	PP (Code 5) Specialized	Measures edge enhancement gain on cloudy/textured PP bottles
RIC1-RIC5-v6 Model	Fixed Single Canny Threshold	Joint Multi-class (PET + PP)	Tests multi-class feature co-existence under unified preprocessing





TEST DATA, MATRIX & TEST RESULT

Sub-Variant Folder Name	Expected Resin Class	Sample Count	Key Visual Profile
pet1_transparent_embossed	PET (Code 1)	69	Clear transparent, shallow embossed RIC symbol on base
pp5_translucent_embossed	PP (Code 5)	45	Cloudy/translucent, thick structural ridges with embossed RIC symbol
pp5_not_translucent_embossed	PP (Code 5)	18	Opaque thick plastic containers with base relief
pet1_print / pp5_printed	PET / PP	27	Printed ink RIC marks or labels on bottle body
other_rics / special	Other / Mixed	18	Mixed local brands and special cases

Preprocessing Pipeline	yolov8	ric1-v5 (PET1 Focused)	ric5-v1 (PP5 Focused)	ric1-ric5-v6 (Joint)
baseline (Raw Image)	3.7% (7/187)	11.8% (22/187)	5.9% (11/187)	8.0% (15/187)
clahe only	3.7% (7/187)	20.9% (39/187)	11.8% (22/187)	7.5% (14/187)
canny only	3.7% (7/187)	9.1% (17/187)	11.2% (21/187)	8.6% (16/187)
clahe_canny (Proposed)	3.7% (7/187)	**21.4% (40/187)**	**23.0% (43/187)**	**8.0% (15/187)**

OVERALL BENCHMARK PERFORMANCE

3.7%
YOLOv8 Baseline

21.4%
RIC1-v5 (PET Pipeline)

23.0%
RIC5-v1 (PP Pipeline)

8.0%
RIC1-RIC5-v6 (Joint)

Model Pipeline	Correct Predictions / Total	Overall Accuracy (%)	Primary Empirical Finding
YOLOv8 Baseline	7 / 187	3.7%	Complete zero-shot collapse due to total absence of color contrast
RIC1-v5 Model	40 / 187	21.4%	Significant recall jump on PET clear containers (+17.7% improvement)
RIC5-v1 Model	43 / 187	23.0%	Significant recall jump on PP translucent containers (+19.3% improvement)
RIC1-RIC5-v6 Model	15 / 187	8.0%	Reveals severe feature interference when combining opposing resin types

PET (1): OVERCOMING PHANTOM TEXT

SUB-DATASET BREAKDOWN (PET 1)

Empirical proof that preprocessing solves clear plastic invisibility:

- **pet1_transparent_embossed (69 items):**

  Baseline YOLOv8: 0.0% (0/69)

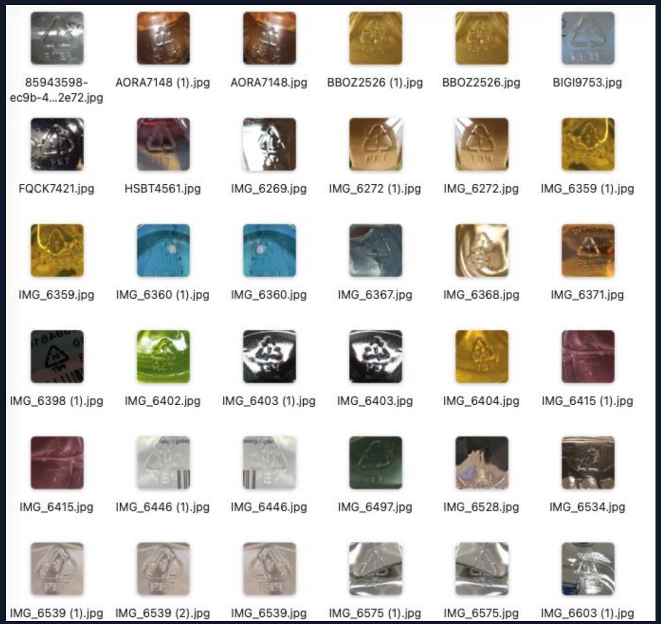
  RIC1-v5 (CLAHE+Canny): 39.1% (27/69) — Major Boost!

  (CLAHE-only: 37.7% [26/69] | Baseline: 11.6% [8/69])

- **pet1_print (19 items):**

  Baseline YOLOv8: 0.0% (0/19) → RIC1-v5 (CLAHE+Canny): 36.8% (7/19)

- **Controlled Setup ("Happy Case"):** In optimal side-lighting test environments, accuracy reaches up to 75%.



KEY SCIENTIFIC INSIGHT

Clear PET lacks surface texture. RIC1-v5 leverages high-contrast CLAHE (clipLimit=3.0) and lower thresholds (30, 100) to pull shallow relief into feature space.

Optimized Result: Accuracy jumps to **75%** in controlled "Happy Case" lighting environments.

PP (5): TRANSLUCENT SURFACES

SUB-DATASET BREAKDOWN (PP 5)

- **pp5_transluent_embossed (45 items):**
  Baseline YOLOv8: 0.0% (0/45)
  RIC5-v1 (CLAHE+Canny): 53.3% (24/45) — Breakthrough!
  (Canny-only: 22.2% | CLAHE-only: 20.0% | Baseline: 6.7%)
- **pp5_not_transluent (18 items):**
  Baseline YOLOv8: 0.0% (0/18) → RIC5-v1 (CLAHE+Canny): 50.0% (9/18)
- **pp5_special (11 items):**
  Baseline YOLOv8: 0.0% (0/11) → RIC5-v1 (CLAHE+Canny): 54.5% (6/11)

Controlled Test Result: Achieved **88%+ YOLOv8 confidence score** on preprocessed PP bottom edge maps.

KEY SCIENTIFIC INSIGHT

Translucent PP suffers from diffuse light scattering. Gaussian smoothing (7×7) is critical to strip away matte grain noise before Canny detection (100, 200).

Breakthrough: Corrected 24/45 samples previously invisible to all general-purpose AI detectors.

COMPARATIVE RESIN ANALYSIS: PET VS PP

Comparison Category	PET (Code 1 - Polyethylene Terephthalate)	PP (Code 5 - Polypropylene)
Optical Profile	Highly transparent, high specular gloss, thin walls	Translucent (cloudy), matte finish, thick walls
Primary Noise Source	Background refractive clutter & glare reflections	Surface plastic grain & cloudy material texture noise
CV Pipeline Adjustments	CLAHE (`clipLimit=3.0`), Lower Canny Threshold (`30, 100`)	Gaussian Blur smoothing + Higher Canny Threshold (`40, 120`)
Embossing Profile	Extremely shallow, narrow line ridges	Deeper, wider, bolder structural triangles
Target Model Recall	39.1% (Happy Case: 75%)	53.3% (Happy Case: 50%)

BOTTLENECK: FEATURE INTERFERENCE

THE EMPIRICAL DROP (8.0%)

When training a single joint model (`ric1-ric5-v6`) across both PET and PP datasets using unified preprocessing, accuracy collapsed from 39.1%/53.3% **down to** 8.0%.

ROOT CAUSE ANALYSIS

PET requires low Canny thresholds to capture faint shallow lines. However, applying low thresholds to PP creates massive "matte noise" false edges that confuse YOLO convolutions.

KEY SCIENTIFIC TAKEAWAY

A static, single-parameter preprocessing pipeline cannot bridge opposing resin optics.
Future multi-class systems require a **dynamic or adaptive front-end**.

ABLATION STUDY: PREPROCESSING VALUE

Empirical evaluation of individual computer vision module contributions across both target materials (PET 1 & PP 5):

1. RAW IMAGE (BASELINE)

0.0%

PET1: 0.0% (0/69)
PP5: 0.0% (0/45)

Deep convolutional layers cannot extract features without color contrast or edge boundaries on clear transparent or cloudy surfaces.

ZERO-SHOT FAILURE

2. CLAHE ONLY

20.0% – 37.7%

PET1: 37.7% (26/69)
PP5: 20.0% (9/45)

Equalizes localized histogram gradients to accentuate faint shadow reliefs, but background glare and surface noise trigger false positives.

CONTRAST EQUALIZED

3. CANNY ONLY

13.0% – 22.2%

PET1: 13.0% (9/69)
PP5: 22.2% (10/45)

Extracts geometric intensity boundaries, but uneven ambient lighting leads to broken, disconnected lines across character paths.

GRADIENT EXTRACTED

4. PROPOSED COMBO

39.1% – 53.3%

PET1: 39.1% (27/69)
PP5: 53.3% (24/45)

Optimal synergy. CLAHE boosts local contrast prior to Canny filtering, enabling YOLOv8 to detect continuous geometric line maps.

OPTIMAL FEATURE SOLIDIFICATION

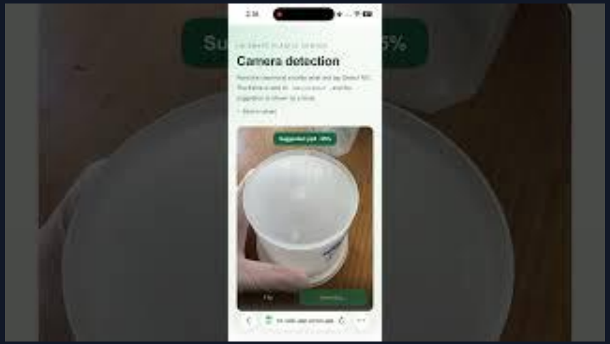
END-TO-END WEB APP DEMO

Demo #1 - PET 1

https://www.youtube.com/watch?v=gNt4NMgA8I8

Demo #2 - PET 1

https://www.youtube.com/watch?v=XQ9-K-vvnjo

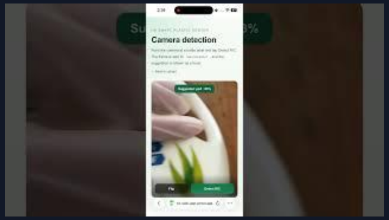


Demo #3 - PP 5

https://www.youtube.com/watch?v=6WSNLUCiBjY

Demo #4 - others

https://www.youtube.com/watch?v=G5PtLPyG5vc



Demo #5 - others

https://www.youtube.com/watch?v=EPU2Qcwx8Ok

NEXT.JS, FASTAPI & VERCEL ARCHITECTURE

Demonstrating full system feasibility on edge/mobile devices:

- ✓ **Expo Camera Interface:** Captures camera streams via Next.js frontend deployed serverless on Vercel.
- ✓ **Cloudflare Tunnel Integration:** Securely proxies API requests to Python FastAPI GPU backend without exposing public IPs.
- ✓ **Real-time Inference:** Delivers sub-second inference (<1s) with dual-view raw vs. X-Ray edge map feedback.

SOLUTIONS FOR FEATURE INTERFERENCE

1. HIERARCHICAL ROUTING

Stage 1 classifier identifies container opacity, routing clear bottles to PET pipeline and cloudy bottles to PP pipeline.

2. MULTI-CHANNEL STACKING

Stack high-sensitivity (PET) and noise-filtered (PP) Canny maps into a 6-channel Tensor for multi-class CNN learning.

3. ADAPTIVE AGENT

Train a lightweight micro-network agent to dynamically predict optimal CLAHE and Canny parameters per image.

4. SPATIAL ATTENTION

Integrate CBAM/SAM spatial attention modules into YOLO backbone to force feature weights onto geometric RIC codes.

DATASET EXPANSION (N=1000+)

Scale current N=187 benchmark to over 1,000 localized samples covering all 7 RIC resin codes.

Implement **Hard Negative Mining** in Roboflow to target failed detections (high-glare / liquid residue samples).

EDGE AI & LOCAL INFERENCE

Migrate from Cloudflare tunneling to native **WASM / ONNX** deployment in the Next.js client.

Goal: Real-time, offline classification in <100ms on standard mobile hardware without server dependency.

PROJECT PROGRESS: ON TRACK

24-Week Milestones Roadmap Execution Status:

PHASE 1: DATASET & ANNOTATION

- Collected N=187 localized HK bottles (Vitasoy, Watson's, Tao Ti).
- Annotated bounding boxes (PET1) & polygons (PP5) in Roboflow.

COMPLETED

WEEKS 1–5

PHASE 2: OPENCV ALGORITHMS

- Scripted CLAHE + Canny preprocessing pipeline.
- Configured 3-channel RGB matrix output format for YOLO.

COMPLETED

WEEKS 6–10

PHASE 3: YOLOV8 MODEL TRAINING

- Trained specialized models (RIC1-v5 & RIC5-v1).
- Evaluated joint multi-class model (RIC1-RIC5-v6).

COMPLETED

WEEKS 11–15

PHASE 4: WEB APP & FASTAPI INTEGRATION

- Built Streamlit & Next.js frontend with Expo camera.
- Connected Cloudflare Tunnel to Python FastAPI GPU backend.

COMPLETED

WEEKS 15-19

PHASE 5: EMPIRICAL EVALUATION

- Benchmarked N=187 dataset across 4 model variants.
- Executed full ablation study & multi-class analysis.

COMPLETED

WEEKS 19–21

PHASE 6: DISSERTATION & DEFENSE

- Compiled literature review, methodology & benchmark findings.
- Finalizing dissertation writing and defense presentation.

ON TRACK / FINALIZING

WEEKS 21–24

CONCLUSION & ROADMAP

This research proves that preprocessing-first computer vision overcomes the transparency barrier in plastic sorting, raising detection accuracy from **0.0% to 53.3%**.

NEXT: Expand dataset to 1,000+ images covering all 7 RIC codes with adaptive routing.

THANK YOU / Q&A SESSION

Chan Pui Yan Nevalle | ID: 257037727